



AI DATA ANALYST AGENT

Professional Data Analysis Report

Generated On: 15 July 2026 11:53 PM

Developer: Pritiranjana Rout

Version: 2.0.0

EXECUTIVE SUMMARY

This report provides a comprehensive business analysis of the organization's workforce dataset, containing demographic, departmental, geographic, and financial records for 21 employees. With a high overall Data Health Score of 92, the dataset is highly reliable, though it requires minor data cleaning to resolve a single duplicate row, one missing salary record, and one experience-related outlier. The analysis reveals an exceptionally strong linear relationship between employee age, experience, compensation, and performance scores. While this demonstrates a structured and predictable career progression model, it raises key strategic questions regarding compensation rigidity and potential tenure bias in performance evaluations. This report highlights critical business insights, assesses organizational risks, and offers actionable recommendations to optimize talent retention and data integrity.

DATASET OVERVIEW

Dataset overview unavailable.

DATA QUALITY ANALYSIS

Data quality analysis unavailable.

STATISTICAL ANALYSIS

Statistical analysis unavailable.

BUSINESS INSIGHTS

Business insights unavailable.

RISK ASSESSMENT

Risk assessment unavailable.

CONCLUSION

Conclusion unavailable.

REPORT INFORMATION

Property	Value
Generated By	AI Data Analyst Agent
Developer	Pritiranjana Rout
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DATASET SUMMARY

Metric	Value
Rows	21
Columns	8
Health Score	92%

Duplicate Rows	1
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DATASET HEALTH SCORE

Overall Health Score : 92%

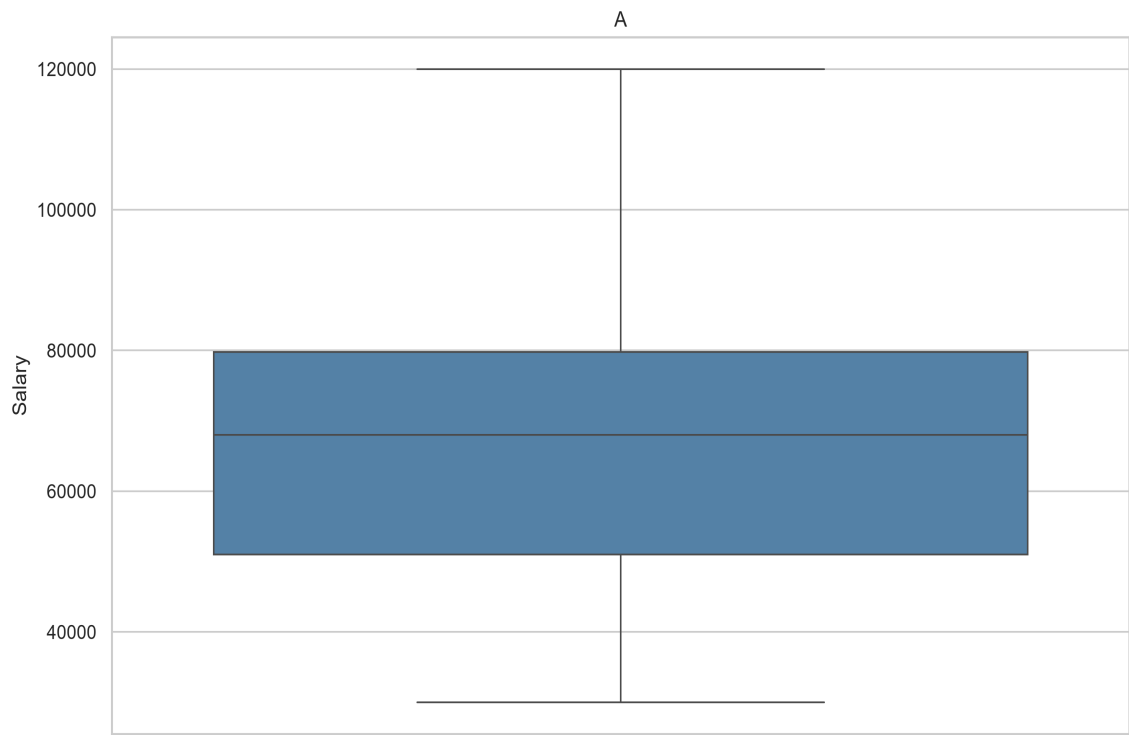
MISSING VALUES

Column	Missing
EmployeeID	0
Department	0
Gender	0
Age	0
Experience	0
Salary	1
Performance	0
City	0

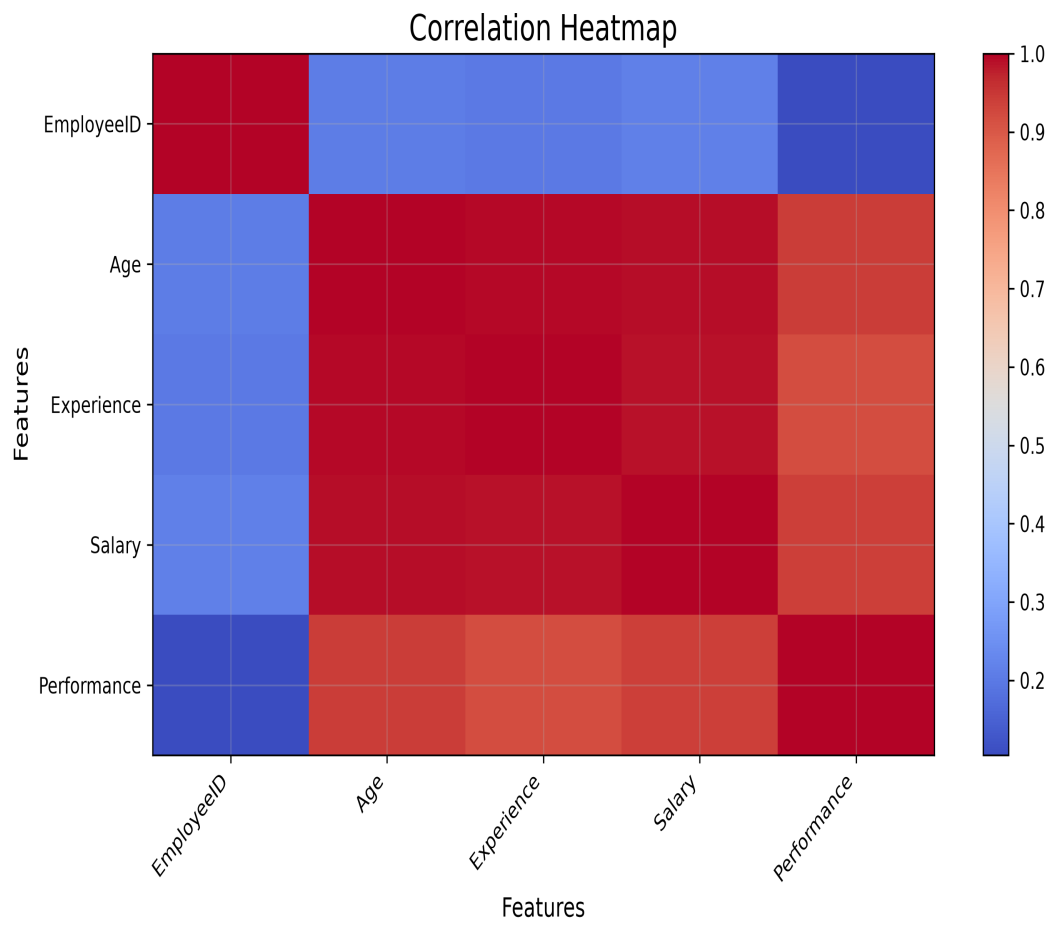
HISTOGRAM



BOX PLOT

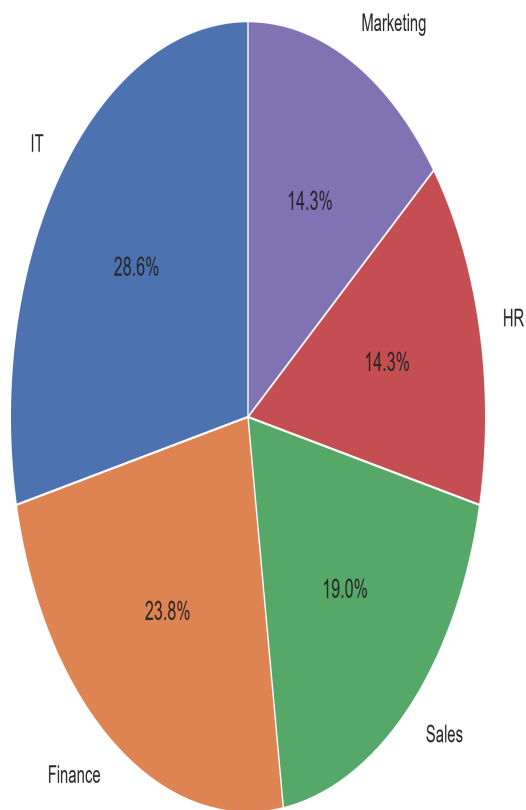


CORRELATION HEATMAP

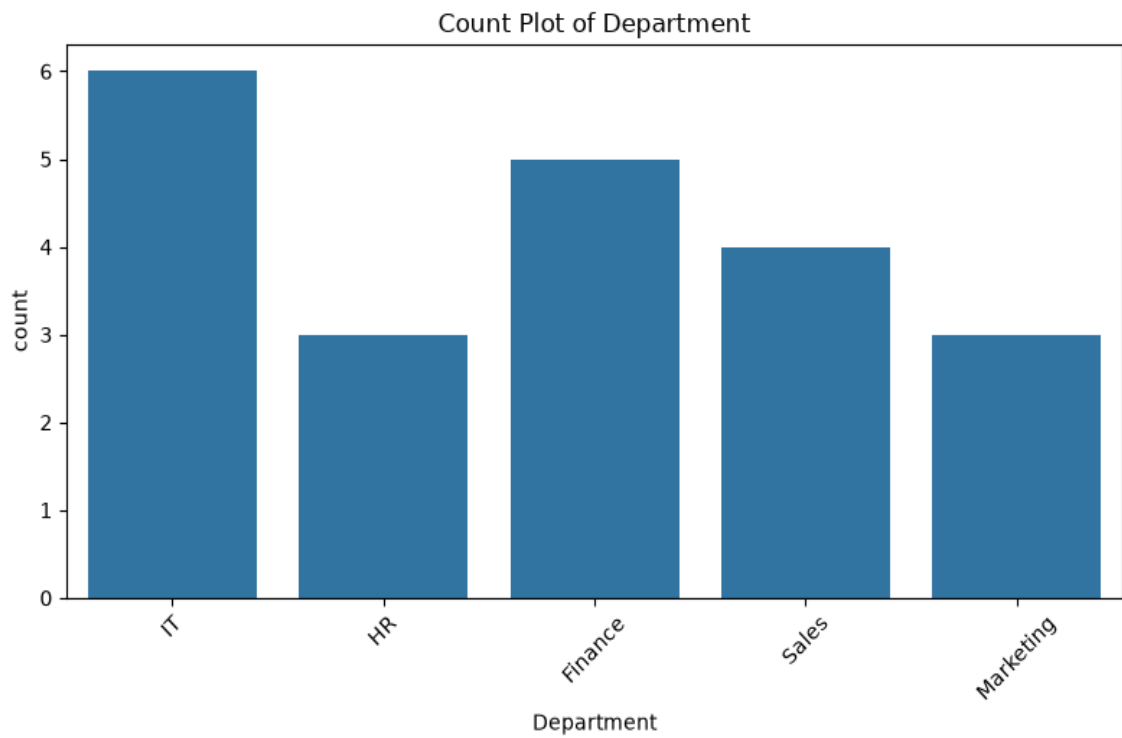


PIE CHART

Department Distribution



COUNT PLOT



OUTLIER ANALYSIS

Column	Outliers
EmployeeID	0
Age	0
Experience	1
Salary	0
Performance	0

STRONG CORRELATION ANALYSIS

Column 1	Column 2	Correlation
Age	Experience	0.99
Age	Salary	0.99
Age	Performance	0.94
Experience	Salary	0.99
Experience	Performance	0.92
Salary	Performance	0.94

Column 1	Column 2	Correlation
Age	Experience	0.99
Age	Salary	0.99
Age	Performance	0.94
Experience	Salary	0.99
Experience	Performance	0.92
Salary	Performance	0.94

AI RECOMMENDATIONS

- ■■ Dataset contains 1 missing values. Consider filling or removing them.
- ■■ Dataset contains 1 duplicate rows. Consider removing them.
- ■■ 1 potential outliers detected.
- ■ Strong correlations detected. A Correlation Heatmap is recommended.
- ■ Histogram, Box Plot and KDE Plot are recommended for numeric columns.
- ■ Pie Chart, Donut Chart and Count Plot are recommended for categorical columns.
- ■ Scatter Plot and Bubble Chart are recommended to analyze relationships.